

TEODORO D. LEITE

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EDUCATION

Boston University, College of Engineering

Boston, MA

B.S. Mechanical Engineering | Concentration: Aerospace Engineering

Expected May 2028

Relevant Coursework: Differential Equations, Engineering Mechanics I, CAD and Machine Components, Electric Circuits, Computational Linear Algebra, Probability & Statistics, Physics I & II

In Progress (Fall 2026): Fluid Mechanics, Energy & Thermodynamics, Mechanics of Materials, Manufacturing Processes, Electromechanical Design

EXPERIENCE

Robosys Automation – Software Development Engineering Intern

Jun 2026 – Present

Machine Learning / Computer Vision | VoyagerAI Maritime Autonomy Program

Team recognized at BlueTIDE 2026 — Collaborative Integration Award and Best Overall On-Water Demonstration

- Develop a camera-based vessel detection and positioning pipeline for a maritime navigation product, written in C#/.NET and targeting an NVIDIA Jetson edge computer
- Train YOLOv8 vessel-detection models in Roboflow across multiple maritime datasets, and build Python tooling to convert, deduplicate, and label tens of thousands of images
- Implement the navigation math layer that converts detections to true bearing and range and outputs NMEA TTM target messages over UDP to an ARPA display
- Wrote a from-scratch AIS decoder in C# (!AIVDM six-bit unpacking, multipart reassembly) and integrated FarSounder forward-looking sonar through its SDK, streaming camera, AIS, and sonar into one target format with per-source ID partitioning; presented the integration plan to the vendor solo
- Design and build an RC data-collection boat — an inflatable dinghy retrofit with dual electric thrusters on differential steering and a 4-camera GMSL rig — and commissioned its SAE J1939 CAN control system solo, configuring PWM outputs to two steering servos and two ESCs with fail-safe neutrals on signal loss

Engineers Without Borders – BU Chapter – Filtration Team Technical Lead

Jan 2026 – Present

Boston, MA

- Lead filtration design for borehole projects serving underserved communities, and run technical learning sessions on filtration methods, fluid systems, and water quality
- Ran the hydraulic analysis for a 150 m borehole serving 1,100 people at Ogiek Secondary School, Keringet, Kenya: 70 m total dynamic head, 11.4 psi gravity delivery at the tap stands, and pump sizing against the manufacturer's curve ($\sim 11 \text{ m}^3/\text{hr}$ at the true operating point) rather than its rated point
- Led the defluoridation media trade study, selecting hydroxylapatite over activated alumina on aluminum-leaching risk, verified gravity feed with a Carman-Kozeny head-loss check (0.37 m required against 9 m available), and produced the SolidWorks elevation, site plan, and flow schematic

Boston University – BUILD STEM Tutor (Federal Work-Study)

Oct 2024 – Present

Boston Public Schools | Boston, MA

- Tutor students in Boston public schools in math and science, plus general homework support, roughly ten hours per week through the academic year
- Work with the same students across the term, teaching to the gap actually blocking them rather than re-reading the assignment

PROJECTS

1984 Pontiac Firebird – Custom Temperature-Controlled Fan System

Jun 2025 – Aug 2025

Personal Project | Fall River, MA

- Designed and built an automated engine cooling system using an LM35 temperature sensor, LM393 comparator IC, and relay module to trigger cooling fans based on real-time engine temperature
- Generated Gerber files and ordered a professionally manufactured PCB through JLCPCB, replacing the breadboard prototype with a durable production-quality board
- Integrated the custom PCB into the Firebird's 12V electrical system with fuse protection, modernizing the car's cooling reliability and reducing unnecessary power draw

Intro to Engineering Design – Mechanical Game Project – Lead Designer

Jan 2026 – Present

Boston University | Boston, MA

- Designed and 3D-printed a custom enclosure for an interactive arcade-style game, modeling all parts and assemblies in SolidWorks and iterating on tolerances and fit across multiple print cycles
- Built the electromechanical system on an Arduino with a stepper motor, joystick, arcade buttons, an LCD display, and audio output, wiring and programming the full control loop

SKILLS

Hardware & Prototyping: Circuit design, PCB fabrication (JLCPCB/Gerber), Soldering (incl. Deutsch DTM & M12 connectors), Embedded systems (Arduino), CAN bus / SAE J1939, Bench debugging, Mechanical design, 3D printing, Machine shop access (BU EPIC)

Software: MATLAB, SolidWorks/CAD, AutoCAD, Python, C++, C#/.NET, WPF/WinForms, NMEA 0183, GeoJSON / ArcGIS REST, Git

Machine Learning: Computer vision, YOLOv8 object detection, Roboflow, PyTorch/ONNX, dataset labeling & augmentation

Navigation & Controls: Target tracking, bearing/range solving, multi-sensor integration, coordinate-frame transforms, dead reckoning, control allocation

Languages: English (fluent), Portuguese (native/bilingual)